Statistics

Exam revision and fully worked solutions

Random variables • Expectation and variance
Binomial • Poisson • One-sample t-tests

Purpose. Expand the supplied 11-page revision notes into a complete examination guide. Each numerical solution explains the model, defines the quantities, substitutes the data, shows the calculation, and ends with an interpretation.

Assignment coverage. All eight questions in Assignment 4, all five in Assignment 5, and the six one-sample questions in Assignment 6 are solved in detail. Assignment 2, Q6(iii), is also included because it is a direct binomial application. A coverage audit appears on page 2.

Practice coverage. The 13 original worked examples are retained within the expanded explanations. A six-question practice paper and complete solutions provide a final rehearsal.

How to use this guide

1. **First pass: understand the method.** Read the theory pages and reproduce the short derivations without looking.
2. **Second pass: solve the assignments.** Cover each solution, write your own setup, then compare the calculation line by line.
3. **Last pass: work under time pressure.** Attempt the practice paper on page 34 and use the formula sheet on page 37 only afterwards.

Exam point. A correct formula earns more credit when the variable, support, parameters, assumptions, and requested event are explicit. Never give an unexplained calculator answer.

Based solely on the seven supplied PDFs. Question statements are restated for clarity; original numerical data are preserved except where a printed error is explicitly identified. No two-sample inference or other additional syllabus units are introduced.

READ THIS BEFORE REVISING

Scope, question map and conventions

Syllabus authority: `Statistics_Midterm_Revision.pdf`. Its scope is random variables; PMFs, PDFs and CDFs; expectation and variance; binomial and Poisson distributions; and one-sample t-tests.

Material	Selection and reason	Guide pages
Assignment 1	Q1–10 excluded: descriptive charts, weighted/geometric means, mode and grouped medians are outside the revision-note units.	—
Assignment 2	Q6(iii) included: two independent draws with replacement. Other questions/subparts concern general counting or conditional probability rather than the specified distribution methods.	13
Assignment 3	Q1–4 excluded: total probability and Bayes' theorem are outside the notes.	—
Assignment 4	Binomial Q1–4 and Poisson Q1–4: all included. Matching Q2 is a general DRV, despite its binomial heading.	14–17; 21–23
Assignment 5	Q1–5: all included, including the table embedded as an image in Q4.	7–11
Assignment 6	Q1, 2, 4, 6, 7, 9 included. Q3, 5, 8, 10, 11 compare two samples and are excluded.	27–32

Source issues handled explicitly

- Assignment 5 Q1 prints $0 \leq 1 \leq x$. This cannot normalize $3x^2$ over $x \geq 1$. The solution uses the intended $0 \leq x \leq 1$ and labels the correction.
- Assignment 6 Q1, Q2 and Q4 omit α. This guide assumes $\alpha = 0.05$; reported p-values let you apply another level.
- Assignment 6 Q1 uses the printed $s = 0.02$ cm, even though it is large relative to the claimed mean. It is not silently changed.
- Assignment 6 Q7 ends with an unexplained ">= 6". This is treated as stray text; $n = 7$ implies six degrees of freedom.

Notation and rounding. X is a random variable, x a possible value, $\mu = \mathrm{E}[X]$, and $\sigma^2 = \mathrm{Var}(X)$. Sample quantities are $\bar{x}, s, n$. Probabilities are rounded at the end, usually to five decimals. Exact fractions are retained when useful. Labels such as A4–B1 and A6–Q2 refer to the original assignment numbering.

FOUNDATIONS

Random variables: from outcomes to numbers

An experiment produces an outcome ω in a sample space Ω. An event is a subset of Ω. A random variable is a function

$$X : \Omega \longrightarrow \mathbb{R}.$$

The function itself is fixed; its observed value is uncertain before the experiment occurs. Different outcomes can produce the same numerical value.

Example: two fair, independent coins

Let X be the number of heads. The four equally likely outcomes are

Outcome	TT	HT	TH	HH
X	0	1	1	2

Thus $P(X = 0) = 1/4$, $P(X = 1) = 1/2$, and $P(X = 2) = 1/4$. The event $\{X = 1\}$ corresponds to two outcomes, not one. Its probability is their combined probability.

Feature	Discrete variable	Continuous variable with a density
Values	Finite or countably infinite	A continuum, often an interval
Description	PMF $p(x) = P(X = x)$	PDF $f(x)$, a density
Validity	$p(x) \geq 0$, $\sum_x p(x) = 1$	$f(x) \geq 0$, $\int f(x)\,dx = 1$
Interval probability	Add included masses	Integrate over the interval
Single point	May have positive mass	$P(X = a) = 0$
Examples	Defects, calls, correct matches	Thickness, lifetime, measured length

The support comes first

The support lists the possible values with positive mass or the interval where a density can be nonzero. If $X \in \{2, 3, 4, 5\}$, then $P(X \leq 2) = P(X = 2)$ and $P(X > 5) = 0$. Writing the support prevents both summation and integration errors.

Three questions to ask before computing

1. What exactly does X measure, and in what units?
2. Which values can occur? Are they isolated counts or continuously varying measurements?
3. Is the supplied expression a probability, a density, or a cumulative probability?

Exam point. "Continuous" does not mean that the density must be a smooth curve everywhere. A piecewise density is allowed. The defining probability calculation is area under the density.

ORIGINAL WORKED EXAMPLE 1

Discrete PMFs and cumulative probabilities

Question. $p(x) = k(x+1)$ for $x = 0, 1, 2$, and zero otherwise. Find k, $P(X \geq 1)$, the mean, variance, standard deviation and CDF.

1. Normalize. Total probability must equal one:

$$k + 2k + 3k = 1 \quad \Rightarrow \quad k = \frac{1}{6}.$$

2. Add the requested masses.

$$P(X \geq 1) = p(1) + p(2) = \frac{2}{6} + \frac{3}{6} = \frac{5}{6}.$$

3. Build a moment table.

x	$p(x)$	$xp(x)$	$x^2p(x)$
0	1/6	0	0
1	2/6	2/6	2/6
2	3/6	6/6	12/6
Total	1	4/3	7/3

$$\mu = \frac{4}{3}, \qquad \sigma^2 = \frac{7}{3} - \left(\frac{4}{3}\right)^2 = \frac{5}{9}, \qquad \sigma = \frac{\sqrt{5}}{3} = 0.74536.$$

4. Accumulate from the left. A CDF is $F(x) = P(X \leq x)$:

$$F(x) = \begin{cases} 0, & x < 0, \\ 1/6, & 0 \leq x < 1, \\ 1/2, & 1 \leq x < 2, \\ 1, & x \geq 2. \end{cases}$$

At $x = 1$, the mass at 1 is already included, so $F(1) = 1/2$. Every CDF is nondecreasing, right-continuous, between 0 and 1, and approaches 0 and 1 at the two extremes.

Endpoint check. $P(0 < X \leq 1) = F(1) - F(0) = 1/2 - 1/6 = 1/3$. In contrast, $P(0 \leq X \leq 1) = 1/2$. For integer-valued X,

$$P(a \leq X \leq b) = F(b) - F(a-1) \quad (a, b \text{ integers}).$$

Exam point. A CDF jump at x equals $P(X = x)$. Draw a step graph with a filled dot at each included left endpoint and an open dot at each excluded right endpoint. The mean 4/3 need not itself be a possible outcome.

ORIGINAL WORKED EXAMPLES 3 AND 4

Expectation, variance and transformations

Expectation is a probability-weighted average. For a DRV,

$$\mathrm{E}[g(X)] = \sum_x g(x)p(x), \quad \mu = \sum_x xp(x), \quad \mathrm{Var}(X) = \mathrm{E}[X^2] - \mu^2.$$

For a continuous variable, replace the sum by $\int g(x)f(x)\,dx$ over the support. The relevant sums or integrals must exist.

Properties worth learning with their conditions

$$\begin{aligned} \mathrm{E}[aX + bY + c] &= a\,\mathrm{E}[X] + b\,\mathrm{E}[Y] + c, \\ \mathrm{Var}(aX + b) &= a^2\,\mathrm{Var}(X), \qquad & \mathrm{SD}(aX + b) &= |a|\,\mathrm{SD}(X), \\ \mathrm{Var}(X + Y) &= \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X, Y). \end{aligned}$$

Linearity of expectation does **not** require independence. For independent variables, $\mathrm{E}[XY] = \mathrm{E}[X]\,\mathrm{E}[Y]$ and $\mathrm{Cov}(X, Y) = 0$. Zero covariance alone need not imply independence. A constant has variance zero.

Why the variance shortcut works:

$$\mathrm{E}[(X - \mu)^2] = \mathrm{E}[X^2] - 2\mu\,\mathrm{E}[X] + \mu^2 = \mathrm{E}[X^2] - \mu^2.$$

Why shifting does not change spread: if $Y = aX + b$, then

$$Y - \mathrm{E}[Y] = a(X - \mu), \quad \mathrm{E}[(Y - \mathrm{E}[Y])^2] = a^2\,\mathrm{Var}(X).$$

Worked example: linear transformation

Given $\mathrm{E}[X] = 4$, $\mathrm{Var}(X) = 9$, and $Y = 3X - 2$,

$$\mathrm{E}[Y] = 3(4) - 2 = 10, \quad \mathrm{Var}(Y) = 3^2(9) = 81, \quad \mathrm{SD}(Y) = 9.$$

If the multiplier were -3, the SD would still be 9. Standard deviation is never negative.

Worked example: nonlinear expectation

If $P(X = 0) = 1/4$ and $P(X = 2) = 3/4$, then

$$\mathrm{E}[X] = 0\left(\tfrac{1}{4}\right) + 2\left(\tfrac{3}{4}\right) = \tfrac{3}{2}, \qquad \mathrm{E}[X^2] = 0 + 4\left(\tfrac{3}{4}\right) = 3.$$

$$\mathrm{E}[2X^2 + 3X + 1] = 2(3) + 3\left(\tfrac{3}{2}\right) + 1 = \frac{23}{2}.$$

Exam point. Generally $\mathrm{E}[g(X)] \neq g(\mathrm{E}[X])$. Here $\mathrm{E}[X^2] = 3$ but $(\mathrm{E}[X])^2 = 9/4$. Mean and SD use the original units; variance uses squared units.

ORIGINAL WORKED EXAMPLE 2

Continuous densities: a complete template

A PDF gives probability through area: $P(a \le X \le b) = \int_a^b f(x)\,dx$. Its height is not a probability and may exceed 1. Endpoints do not affect interval probabilities because $P(X = a) = 0$.

Question. $f(x) = kx$ on $[0, 1]$, zero otherwise. Find k, $P(1/4 \le X \le 3/4)$, the moments and CDF.

$$1 = \int_0^1 kx\,dx = k\left[\frac{x^2}{2}\right]_0^1 = \frac{k}{2} \qquad \Rightarrow k = 2,$$

$$P(1/4 \le X \le 3/4) = \int_{1/4}^{3/4} 2x\,dx = \Big[x^2\Big]_{1/4}^{3/4} = \frac{9-1}{16} = \frac{1}{2}.$$

Multiplying the density by x gives the first moment; multiplying by x^2 gives the second:

$$\mu = \int_0^1 x(2x)\,dx = 2\left[\frac{x^3}{3}\right]_0^1 = \frac{2}{3},$$

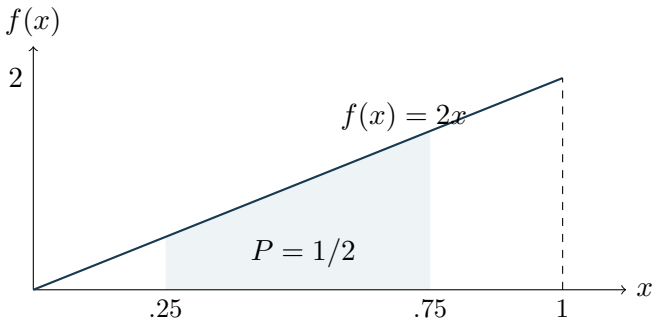
$$\mathrm{E}[X^2] = \int_0^1 x^2(2x)\,dx = 2\left[\frac{x^4}{4}\right]_0^1 = \frac{1}{2},$$

$$\sigma^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}, \qquad \sigma = 0.23570.$$

The cumulative probability must also be defined outside the support:

$$F(x) = \begin{cases} 0, & x < 0, \\ \int_0^x 2u\,du = x^2, & 0 \le x \le 1, \\ 1, & x > 1. \end{cases}$$

The letter u is a dummy variable; using it keeps the upper limit x distinct from the integration variable. On the interior, $F'(x) = 2x = f(x)$.



Exam point. Use $\int x^m dx = x^{m+1}/(m+1) + C$ for $m \neq -1$. In a definite integral, evaluate the antiderivative at the upper limit and subtract its value at the lower limit.

ASSIGNMENT 5 / RANDOM VARIABLES

A5–Q1: split a density into equal halves

Source: Assignment 5, Q1.

Question. For the density $f(x) = 3x^2$, find a such that $P(X \le a) = P(X > a)$.

Printed-support correction. The assignment states $0 \le 1 \le x$. On $[1, \infty)$, $\int_1^\infty 3x^2 dx$ diverges, so this cannot be a PDF. We use the intended support $0 \le x \le 1$, with density zero elsewhere. This correction is necessary for the problem to have the standard answer.

Step 1: check that the corrected density is valid

It is nonnegative on $[0, 1]$, and

$$\int_0^1 3x^2 \, dx = \Big[x^3\Big]_0^1 = 1.$$

Step 2: translate "equal probability"

The two events partition the whole sample space. Their probabilities sum to one. If they are equal, each must equal one half:

$$P(X \le a) = P(X > a) = \frac{1}{2}.$$

This is the definition of a median. It is not a request to compute the mean.

Step 3: form the CDF and solve

For $0 \le a \le 1$,

$$P(X \le a) = \int_0^a 3x^2 \, dx = a^3.$$

Consequently,

$$a^3 = \frac{1}{2} \quad \Rightarrow \quad \boxed{a = 2^{-1/3} = 0.79370}.$$

Step 4: verify the other half

$$P(X > a) = \int_a^1 3x^2 \, dx = 1 - a^3 = 1 - \frac{1}{2} = \frac{1}{2}.$$

The result lies inside the support, as it must. The median is greater than 0.5 because the increasing density places more probability near the right endpoint.

Understand the distinction from an average

As a useful check on concepts, the mean would be

$$\mathrm{E}[X] = \int_0^1 x(3x^2) \, dx = 3\left[\frac{x^4}{4}\right]_0^1 = \frac{3}{4}.$$

Thus the mean is 0.75, while the equal-area cutoff is approximately 0.79370. Neither calculation requires probabilities at individual points.

Exam point. For an equal-tail cutoff in a continuous distribution, solve $F(a) = 1/2$. More generally, a cumulative-probability cutoff u satisfies $F(a) = u$.

ASSIGNMENT 5 / RANDOM VARIABLES

A5–Q2: normalize before integrating

Source: Assignment 5, Q2.

Question. X takes values between 2 and 5 and has PDF $f(x) = k(1+x)$. Find $P(X < 4)$.

Step 1: specify the full density

$$f(x) = \begin{cases} k(1+x), & 2 \le x \le 5, \\ 0, & \text{otherwise.} \end{cases}$$

The constant k controls the total area. It must be found before an interval integral can be interpreted as a probability.

Step 2: set total area equal to one

$$\begin{aligned} 1 &= \int_2^5 k(1+x)\,dx = k\left[x + \frac{x^2}{2}\right]_2^5 \\ &= k\left(5 + \frac{25}{2} - 2 - \frac{4}{2}\right) = k\left(\frac{35}{2} - 4\right) = \frac{27k}{2}. \end{aligned}$$

$$\boxed{k = \frac{2}{27}}$$

The resulting density is positive throughout its support.

Step 3: integrate only over the requested event

Because values below 2 are impossible,

$$\begin{aligned} P(X < 4) &= \int_2^4 \frac{2}{27}(1+x)\,dx \\ &= \frac{2}{27}\left[x + \frac{x^2}{2}\right]_2^4 = \frac{2}{27}\left[(4+8) - (2+2)\right] \\ &= \boxed{\frac{16}{27} = 0.59259}. \end{aligned}$$

Using $X \le 4$ gives the same answer because $P(X = 4) = 0$.

CDF check and interpretation

The corresponding CDF, useful for checking your integration, is

$$F(x) = \begin{cases} 0, & x < 2, \\ (x^2 + 2x - 8)/27, & 2 \le x \le 5, \\ 1, & x > 5. \end{cases}$$

It satisfies $F(2) = 0$, $F(5) = 1$ and $F(4) = 16/27$. Its derivative on $(2, 5)$ is $(2x+2)/27$, exactly the density.

Approximately 59.259% of the distribution lies below 4. The complementary probability is $P(X \ge 4) = 11/27$; the two add to one.

Exam point. Do not integrate from 0 just because a previous question did. The lower integration limit is the lower endpoint of this question's support.

ASSIGNMENT 5 / RANDOM VARIABLES

A5–Q3: inverse-power density and CDF

Source: Assignment 5, Q3.

Question. $f(x) = k/x^5$ for $2 \le x \le 4$, and zero otherwise. Find k, the CDF, and $P(2.5 \le X \le 3)$.

1. Normalizing constant

Write $1/x^5 = x^{-5}$. Its antiderivative is $-x^{-4}/4$:

$$\begin{aligned} 1 &= k \int_2^4 x^{-5}\,dx = k\left[-\frac{1}{4x^4}\right]_2^4 \\ &= \frac{k}{4}\left(\frac{1}{16} - \frac{1}{256}\right) = \frac{15k}{1024}. \end{aligned}$$

$$\boxed{k = \frac{1024}{15}}$$

A constant greater than one is allowed: k is not itself a probability.

2. Full distribution function

For an interior cutoff x,

$$\begin{aligned} F(x) &= \int_2^x \frac{1024}{15u^5}\,du = \frac{256}{15}\left(\frac{1}{16} - \frac{1}{x^4}\right) \\ &= \frac{16}{15} - \frac{256}{15x^4}. \end{aligned}$$

Hence

$$\boxed{F(x) = \begin{cases} 0, & x < 2, \\ \dfrac{16}{15} - \dfrac{256}{15x^4}, & 2 \le x \le 4, \\ 1, & x > 4. \end{cases}}$$

Check: $F(2) = 0$, $F(4) = 1$, and $F'(x) = 1024/(15x^5) > 0$ on the support. These checks catch sign mistakes in the antiderivative.

3. Requested interval probability

$$\begin{aligned} P(2.5 \le X \le 3) &= F(3) - F(2.5) \\ &= \frac{256}{15}\left(\frac{1}{(2.5)^4} - \frac{1}{3^4}\right) \\ &= \frac{256}{15}\left(\frac{16}{625} - \frac{1}{81}\right) = \boxed{\frac{171776}{759375} = 0.22621}. \end{aligned}$$

Exam point. Keep the factor $1/4$ from integration. Using k instead of $k/4$ in the CDF multiplies every interval answer by four.

ASSIGNMENT 5 / RANDOM VARIABLES

A5–Q4: blemished tires and moments

Source: Assignment 5, Q4; probabilities transcribed from the embedded table.

Question. The daily count X of sound but blemished tires has the following distribution. Find (a) $P(X > 3)$, (b) $P(X \le 2)$, and (c) the mean and standard deviation.

x	$p(x)$	$xp(x)$	$x^2p(x)$
2	0.48	0.96	1.92
3	0.36	1.08	3.24
4	0.12	0.48	1.92
5	0.04	0.20	1.00
Total	1.00	2.72	8.08

All masses are nonnegative and their sum is one, so this is a valid PMF.

(a) More than three

The event contains only the possible values 4 and 5:

$$P(X > 3) = P(X = 4) + P(X = 5) = 0.12 + 0.04 = \boxed{0.16}.$$

Thus there is a 16% probability of producing more than three blemished tires tomorrow, assuming tomorrow follows this daily distribution.

(b) At most two

The smallest possible value is 2. There are no masses at 0 or 1:

$$P(X \le 2) = P(X = 2) = \boxed{0.48}.$$

(c) Mean and standard deviation

Read the last two table totals as moments:

$$\mu = \sum_x xp(x) = 2.72, \qquad \mathrm{E}[X^2] = \sum_x x^2p(x) = 8.08.$$

$$\begin{aligned} \sigma^2 &= \mathrm{E}[X^2] - \mu^2 = 8.08 - (2.72)^2 \\ &= 8.08 - 7.3984 = 0.6816, \\ \sigma &= \sqrt{0.6816} = \boxed{0.82559 \text{ tires}}. \end{aligned}$$

Interpretation. The long-run mean is $\boxed{2.72}$ blemished tires per day; it need not be an integer. The variance is 0.6816 tires squared.

Optional CDF check. Successive cumulative masses are $0.48, 0.84, 0.96, 1$ at $2, 3, 4, 5$. Therefore $P(X > 3) = 1 - F(3) = 1 - 0.84 = 0.16$, agreeing with the direct sum.

Exam point. "More than three" excludes 3. "At most two" includes 2. Translate the wording into an inequality before looking at the table.

ASSIGNMENT 5 / RANDOM VARIABLES

A5–Q5: lottery net gain and risk

Source: Assignment 5, Q5.

Question. There are 5,000 tickets at \$1 each. One wins \$1,000, two win \$500 each, and ten win \$100 each. For one randomly selected ticket, construct the distribution of net gain X, and find its mean and SD.

1. Construct the distribution

There are $5000 - (1 + 2 + 10) = 4987$ nonwinning tickets. Net gain equals prize minus the \$1 cost, including on a losing ticket.

Prize	Net gain x	Tickets	$p(x)$
\$0	-1	4987	4987/5000
\$100	99	10	10/5000
\$500	499	2	2/5000
\$1000	999	1	1/5000

The probabilities sum to one because the ticket counts sum to 5,000.

2. Expected net gain

$$\begin{aligned}\mathrm{E}[X] &= \frac{-4987 + 99(10) + 499(2) + 999(1)}{5000} \\ &= \frac{-2000}{5000} = \boxed{-\$0.40}.\end{aligned}$$

A ticket produces an average loss of 40 cents over repeated comparable purchases. This does not say that a single ticket can have an outcome of $-\$0.40$.

3. Second moment, variance and SD

$$\begin{aligned}\mathrm{E}[X^2] &= \frac{(-1)^2(4987) + 99^2(10) + 499^2(2) + 999^2}{5000} \\ &= \frac{4987 + 98010 + 498002 + 998001}{5000} = 319.8, \\ \mathrm{Var}(X) &= 319.8 - (-0.4)^2 = 319.64, \\ \mathrm{SD}(X) &= \sqrt{319.64} = \boxed{\$17.87848}.\end{aligned}$$

Independent arithmetic check using gross prize

Let $W = X + 1$. Then $\mathrm{E}[W] = 3000/5000 = 0.6$ and

$$\mathrm{E}[W^2] = \frac{1000^2 + 2(500^2) + 10(100^2)}{5000} = 320.$$

Thus $\mathrm{Var}(W) = 320 - 0.6^2 = 319.64 = \mathrm{Var}(X)$, because subtracting a constant changes the mean but leaves variance unchanged.

Exam point. Large rare prizes can produce a large SD even when almost every ticket loses. Do not confuse the average loss with the variability of individual outcomes.

THEORY AND MODEL SELECTION

Binomial distribution: recognize and derive

$X \sim \mathrm{Bin}(n, p)$ counts successes in n trials. "Success" is simply the event being counted; it can mean a defect or a failed attempt.

1. The number of trials n is fixed in advance.
2. Each trial has two categories: success and failure.
3. Trials are independent.
4. The success probability p is the same on every trial.

Write $q = 1 - p$. The support is $0, 1, \ldots, n$ and

$$\boxed{P(X = r) = \binom{n}{r} p^r q^{n-r}}, \qquad \binom{n}{r} = \frac{n!}{r!(n-r)!}, \quad 0! = 1.$$

Why the formula contains a combination

A particular ordered sequence with r successes and $n - r$ failures has probability $p^r q^{n-r}$ by independence. There are $\binom{n}{r}$ such sequences. They are mutually exclusive, so their probabilities add.

For three trials, exactly two successes can occur as SSF, SFS or FSS. Each has probability $p^2 q$, giving $3p^2q = \binom{3}{2}p^2q$. Writing only p^2q counts one ordering and misses the other two.

Mean and variance from indicator variables

Let $I_i = 1$ if trial i succeeds and 0 otherwise. Since $I_i^2 = I_i$,

$$\mathrm{E}[I_i] = p, \quad \mathrm{E}[I_i^2] = p, \quad \mathrm{Var}(I_i) = p - p^2 = pq.$$

Now $X = I_1 + \cdots + I_n$. Linearity and independence respectively give

$$\boxed{\mathrm{E}[X] = np}, \qquad \boxed{\mathrm{Var}(X) = npq}, \qquad \boxed{\mathrm{SD}(X) = \sqrt{npq}}.$$

Shape and practical checks

For $0 < p < 1$, the distribution is symmetric when $p = 1/2$; small p produces a longer right tail and large p a longer left tail. When $p = 0$ or $p = 1$, all probability is concentrated at one endpoint.

Check that $0 \le p \le 1$, n is a nonnegative integer, and the requested count is in the support. The variance cannot exceed the mean because $npq \le np$.

Exam point. Sampling without replacement generally breaks independence and changes the success probability. A matching problem may have a fixed number of answers and two outcomes per answer yet still fail the binomial conditions.

ORIGINAL EXAMPLE 5 + ASSIGNMENT 2 Q6(iii)

Binomial wording and short applications

For integer k and a count X, translate the event before substituting:

Wording	Probability calculation
Exactly k	$P(X = k)$
At most k	$\sum_{r=0}^{k} P(X = r)$
Fewer than k	$\sum_{r=0}^{k-1} P(X = r)$
At least k	$1 - \sum_{r=0}^{k-1} P(X = r)$
More than k	$1 - \sum_{r=0}^{k} P(X = r)$
At least one	$1 - P(X = 0) = 1 - q^n$

Worked example: five items, defect probability 0.4

Let X count defects among five independent items. Then $X \sim \text{Bin}(5, 0.4)$, with $q = 0.6$.

$$
\begin{aligned}
P(X = 2) &= \binom{5}{2}(0.4)^2(0.6)^3 = 10(0.16)(0.216) = \boxed{0.34560}, \\
P(X \le 1) &= (0.6)^5 + 5(0.4)(0.6)^4 \\
&= 0.07776 + 0.25920 = \boxed{0.33696}, \\
P(X \ge 1) &= 1 - (0.6)^5 = \boxed{0.92224}.
\end{aligned}
$$

Also $\text{E}[X] = 5(0.4) = 2$ and $\text{Var}(X) = 5(0.4)(0.6) = 1.2$. Notice that the complement of "at most one" is "at least two", not "at least one".

A2–Q6(iii): two gloves drawn with replacement

Source: Assignment 2, Q6(iii).

Question. A box contains seven right-handed and two left-handed gloves. Two are randomly selected with replacement. Find the probability that both are right-handed.

Let X count right-handed selections. Replacement restores the original composition before the second independent draw, so $n = 2$, $p = 7/9$, $q = 2/9$ and $X \sim \text{Bin}(2, 7/9)$.

$$
P(X = 2) = \binom{2}{2}\left(\frac{7}{9}\right)^2\left(\frac{2}{9}\right)^0 = \boxed{\frac{49}{81} = 0.60494}.
$$

The multiplication-rule check is $(7/9)(7/9)$, the same result. This answer applies specifically to the with-replacement subpart; the assignment's other glove draws are without replacement.

Exam point. For a discrete count, moving from $X \ge k$ to its complement changes the cutoff to $X \le k - 1$. That one-unit adjustment matters.

ASSIGNMENT 4 / BINOMIAL Q1

A4–B1: ten fair coins, all six parts

Source: Assignment 4, Binomial Q1.

Question. Ten unbiased coins are tossed. Find the probabilities of exactly six heads, at least eight heads, no head, at least one head, not more than three heads, and at least four heads.

Assume independent toss outcomes. Let X count heads. Then

$$X \sim \mathrm{Bin}(10, 1/2), \qquad P(X = r) = \frac{\binom{10}{r}}{2^{10}} = \frac{\binom{10}{r}}{1024}.$$

The coefficients for $r = 0, 1, \ldots, 10$ are

$$1,\ 10,\ 45,\ 120,\ 210,\ 252,\ 210,\ 120,\ 45,\ 10,\ 1.$$

They sum to 1024; symmetry reflects the equal head and tail probabilities.

(i) Exactly six heads

$$P(X = 6) = \frac{\binom{10}{6}}{1024} = \boxed{\frac{210}{1024} = 0.20508}.$$

(ii) At least eight heads

Include 8, 9 and 10:

$$P(X \geq 8) = \frac{45 + 10 + 1}{1024} = \boxed{\frac{56}{1024} = 0.05469}.$$

(iii) No head

$$P(X = 0) = \left(\frac{1}{2}\right)^{10} = \boxed{\frac{1}{1024} = 0.00098}.$$

(iv) At least one head

$$P(X \geq 1) = 1 - P(X = 0) = \boxed{\frac{1023}{1024} = 0.99902}.$$

(v) Not more than three heads

$$P(X \leq 3) = \frac{1 + 10 + 45 + 120}{1024} = \boxed{\frac{176}{1024} = 0.17188}.$$

(vi) At least four heads

$$P(X \geq 4) = 1 - P(X \leq 3) = \boxed{\frac{848}{1024} = 0.82813}.$$

Exam point. Parts (v) and (vi) are complements and must sum to one. For a fair-coin model, $P(X = 6) = P(X = 4)$ supplies another immediate check.

ASSIGNMENT 4 / A DRV THAT IS NOT BINOMIAL

A4–B2: correct historical matches

Source: Assignment 4, Binomial Q2.

Question. A student matches Gandhi's birth, India's freedom and the First World War to 1947, 1914 and 1869 by guessing. Obtain the distribution of the number of correct answers.

Interpretation. In a matching exercise each year is used exactly once. All $3! = 6$ one-to-one assignments are assumed equally likely. The correct answers, in the stated event order, are $(1869, 1947, 1914)$. Represent those correct years by $(1, 2, 3)$.

1. Enumerate the equally likely assignments

A correct answer is a number in its original position.

Assignment	Correct positions	Count X
$(1,2,3)$	All three	3
$(1,3,2)$	First	1
$(2,1,3)$	Third	1
$(2,3,1)$	None	0
$(3,1,2)$	None	0
$(3,2,1)$	Second	1

2. Divide each count of arrangements by six

x	0	1	2	3
$P(X=x)$	$2/6 = 1/3$	$3/6 = 1/2$	0	$1/6$

Exactly two correct matches are impossible: if two events receive their correct years, the only remaining year is also correct for the third.

3. Why the binomial formula fails

Each event individually has probability $1/3$ of a correct match, but the correctness indicators are dependent. Their sum therefore does not follow $\text{Bin}(3, 1/3)$. The assignment's section heading does not override the model assumptions.

As a moment check, $\text{E}[X] = 0(1/3) + 1(1/2) + 3(1/6) = 1$, while $\text{E}[X^2] = 1/2 + 9/6 = 2$, so $\text{Var}(X) = 1$. The binomial variance would instead be $2/3$.

Exam point. If the problem explicitly allowed independent answers with repeated years, the model would become $\text{Bin}(3, 1/3)$ with probabilities $8/27, 12/27, 6/27, 1/27$. That is a different guessing procedure.

ASSIGNMENT 4 / BINOMIAL Q3(a,b)

A4–B3: fit eight-coin frequencies

Source: Assignment 4, Binomial Q3.

Question. Eight coins are tossed together 256 times. Find the expected frequencies of the possible head counts, then the mean and SD of the theoretical distribution.

Model assumption. The usual intended model is eight fair coins with independent outcomes in each batch. The assignment does not explicitly say "fair" in this question, so this assumption is stated rather than hidden.

1. Separate the two counts in the question

There are $n = 8$ trials in one batch and $N = 256$ repetitions of the entire experiment. Let X be heads in one batch:

$$X \sim \mathrm{Bin}(8, 1/2), \qquad P(X = r) = \binom{8}{r}\left(\frac{1}{2}\right)^8 = \frac{\binom{8}{r}}{256}.$$

Expected frequency equals number of repetitions times probability:

$$f_r^{\exp} = NP(X = r) = 256\frac{\binom{8}{r}}{256} = \binom{8}{r}.$$

2. Tabulate every possible outcome

Heads r	Probability	Expected frequency
0	1/256	1
1	8/256	8
2	28/256	28
3	56/256	56
4	70/256	70
5	56/256	56
6	28/256	28
7	8/256	8
8	1/256	1
Total	1	256

3. Mean and standard deviation per batch

$$\mu = np = 8\left(\frac{1}{2}\right) = \boxed{4}, \qquad \sigma^2 = npq = 8\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = 2,$$

$$\boxed{\sigma = \sqrt{2} = 1.41421 \text{ heads per batch}}.$$

Exam point. The expected total heads over all batches would be $256 \times 4 = 1024$. That is a different quantity from the requested distribution mean of four heads per batch. Expected frequencies describe averages, not guaranteed observed counts.

ASSIGNMENT 4 / BINOMIAL Q4

A4–B4: six dice, 729 repetitions

Source: Assignment 4, Binomial Q4.

Question. Six dice are thrown 729 times. How many times do you expect at least three dice to show a five or a six?

1. Define success for one die

Assume fair independent dice. A success is either a five or a six, so

$$p = \frac{2}{6} = \frac{1}{3}, \qquad q = \frac{2}{3}.$$

Let X be the number of successful dice in one six-die throw. Then $X \sim \mathrm{Bin}(6, 1/3)$. The number 729 is the repetition count N, not the binomial trial count n.

2. Translate and calculate the required probability

"At least three" means $X = 3, 4, 5, 6$. Thus

$$\begin{aligned} P(X \geq 3) &= \sum_{r=3}^{6} \binom{6}{r} \left(\frac{1}{3}\right)^r \left(\frac{2}{3}\right)^{6-r} \\ &= \frac{1}{3^6} \sum_{r=3}^{6} \binom{6}{r} 2^{6-r}. \end{aligned}$$

r	$\binom{6}{r}$	2^{6-r}	Numerator contribution
3	20	8	160
4	15	4	60
5	6	2	12
6	1	1	1
Total			233

Therefore

$$P(X \geq 3) = \frac{233}{729} = 0.31962.$$

3. Convert probability into expected frequency

$$\text{Expected qualifying throws} = NP(X \geq 3) = 729 \left(\frac{233}{729}\right) = \boxed{233}.$$

This is an expected number, not a certainty about the experimental outcome.

4. Complement check

The excluded counts are 0, 1 and 2. Their numerator is

$$2^6 + 6(2^5) + 15(2^4) = 64 + 192 + 240 = 496.$$

Thus the desired numerator is $729 - 496 = 233$, agreeing with the direct calculation.

Exam point. The probability of at least three successes is not $\binom{6}{3}p^3q^3$ alone. That term counts exactly three and omits four, five and six.

ORIGINAL WORKED EXAMPLES 6 AND 7

Binomial inverse and minimum-trial problems

Example 6: recover parameters from moments

Question. A binomial random variable has mean 4 and variance 3. Find n and p.

From $\mu = np$ and $\sigma^2 = np(1-p)$, dividing the second equation by the first gives, for $\mu > 0$,

$$q = 1 - p = \frac{\sigma^2}{\mu}.$$

Substitute the given values:

$$q = \frac{3}{4}, \qquad p = 1 - \frac{3}{4} = \frac{1}{4}, \qquad n = \frac{\mu}{p} = \frac{4}{1/4} = \boxed{16}.$$

Thus $\boxed{X \sim \mathrm{Bin}(16, 1/4)}$. Check both conditions:

$$16\left(\frac{1}{4}\right) = 4, \qquad 16\left(\frac{1}{4}\right)\left(\frac{3}{4}\right) = 3.$$

The reconstructed n must be an integer and $0 \le p \le 1$. For a positive-mean nondegenerate finite binomial model, $0 < \sigma^2 < \mu$. Do not force impossible moments into the model. Zero mean or zero variance requires examining the degenerate cases separately.

Example 7: minimum attempts for a target probability

Question. Independent attempts succeed with probability 0.2. Find the smallest n for which the chance of at least one success is at least 0.95.

Here $q = 0.8$. Use the zero-success complement:

$$\begin{aligned} P(X \ge 1) \ge 0.95 &\iff 1 - (0.8)^n \ge 0.95 \\ &\iff (0.8)^n \le 0.05 \\ &\iff n \ln(0.8) \le \ln(0.05). \end{aligned}$$

Since $\ln(0.8) < 0$, dividing reverses the inequality:

$$n \ge \frac{\ln(0.05)}{\ln(0.8)} = 13.42513\ldots \quad \Rightarrow \quad \boxed{n_{\min} = 14}.$$

Verify minimality:

$$1 - (0.8)^{13} = 0.94502 < 0.95, \qquad 1 - (0.8)^{14} = 0.95602 \ge 0.95.$$

Checking both integers proves that 14 works and 13 does not.

Exam point. For $0 < p < 1$ and target $0 < c < 1$, the minimum is $\lceil \ln(1-c)/\ln(1-p) \rceil$. Round upward, not to the nearest integer, and verify the adjacent integer.

THEORY + ORIGINAL WORKED EXAMPLE 8

Poisson distribution: counts in an interval

$X \sim \text{Pois}(\lambda)$ models a count in a specified interval, where λ is the expected count in that interval. For $\lambda > 0$,

$$\boxed{P(X = r) = \frac{e^{-\lambda}\lambda^r}{r!}}, \qquad r = 0, 1, 2, \ldots.$$

There is no fixed finite maximum. If $\lambda = 0$, all mass is at zero. Unlike binomial p, λ is not a probability and can exceed one.

Model conditions and rate conversion

The usual model assumes a constant event rate, independent counts in disjoint intervals, and negligible probability of multiple events in a sufficiently short interval relative to one event. Clustering or a changing rate can invalidate the model.

$$\boxed{\lambda = \rho L},$$

where ρ is the rate and L is the interval size in matching units. Always identify the interval to which the given mean applies.

Mean, variance and a useful recurrence

$$\text{E}[X] = \lambda, \quad \text{Var}(X) = \lambda, \quad \text{SD}(X) = \sqrt{\lambda}.$$

For example, cancelling $r/r!$ in the expectation sum gives

$$\text{E}[X] = e^{-\lambda} \sum_{r=1}^{\infty} \frac{r\lambda^r}{r!} = \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} = \lambda.$$

Similarly $\text{E}[X(X-1)] = \lambda^2$, so $\text{E}[X^2] = \lambda^2 + \lambda$, and subtracting λ^2 gives the variance. Consecutive probabilities satisfy

$$P_0 = e^{-\lambda}, \qquad \boxed{P_{r+1} = \frac{\lambda}{r+1} P_r}.$$

This avoids repeated factorial calculations when building a table.

Example 8: three calls per minute

Let X be calls in one minute, so $\lambda = 3$.

$$P(X = 2) = e^{-3}\frac{3^2}{2!} = \boxed{0.22404},$$
$$P(X \le 1) = e^{-3}(1 + 3) = \boxed{0.19915},$$
$$P(X \ge 2) = 1 - e^{-3}(1 + 3) = \boxed{0.80085}.$$

The mean and variance are both 3; the SD is $\sqrt{3}$. The equality of mean and variance is a feature of the model, not a property of arbitrary count data.

Exam point. A Poisson probability plot consists of isolated masses at integer counts. A smooth bell curve is not the PMF, even when the distribution becomes nearly symmetric for larger λ.

ORIGINAL WORKED EXAMPLES 9 AND 10

Poisson intervals and rare-event approximation

Example 9: convert hours to a quarter-hour

Question. Arrivals occur according to a Poisson model at six per hour. Find the probabilities of zero and at least two arrivals in 15 minutes.

Let X count arrivals in the requested 15 minutes. Convert the duration to hours before calculating the mean:

$$L = \frac{15}{60} = \frac{1}{4} \text{ hour}, \qquad \lambda = \rho L = 6\left(\frac{1}{4}\right) = 1.5.$$

$$\begin{aligned} P(X = 0) &= e^{-1.5}\frac{1.5^0}{0!} = \boxed{0.22313}, \\ P(X \geq 2) &= 1 - [P(X = 0) + P(X = 1)] \\ &= 1 - e^{-1.5}(1 + 1.5) = \boxed{0.44217}. \end{aligned}$$

Using $\lambda = 6$ would answer a one-hour question, not this quarter-hour question. The probability does not scale linearly with time; the expected count does.

Example 10: approximate a binomial rare-event count

Question. Each of 100 independent components is faulty with probability 0.02. Use Poisson to approximate the probability of exactly two faulty components.

The exact model is $X \sim \text{Bin}(100, 0.02)$. With many trials and small success probability, use $Y \sim \text{Pois}(np)$:

$$\lambda = np = 100(0.02) = 2.$$

$$P(X = 2) \approx P(Y = 2) = e^{-2}\frac{2^2}{2!} = \boxed{0.27067}.$$

The exact check is

$$P(X = 2) = \binom{100}{2}(0.02)^2(0.98)^{98} = 0.27341.$$

The absolute difference is about 0.00274. Write an approximation sign when using a different distribution.

Feature	Binomial	Poisson
Fixed quantity	Number of trials n	Interval and mean λ
Parameters	n, p	λ
Support	$0, \ldots, n$	$0, 1, 2, \ldots$
Mean / variance	np / $np(1-p)$	λ / λ

Exam point. The limiting approximation has $n \to \infty$, $p \to 0$, and $np \to \lambda$. There is no universal cutoff guaranteeing accuracy for every event. Use it when requested or justified by a rare-event model.

ASSIGNMENT 4 / POISSON Q1

A4–P1: infer the Poisson parameter

Source: Assignment 4, Poisson Q1.

Question. A Poisson variable has mean λ. Given $P(X = 8) = P(X = 9)$, determine $P(4 < X \leq 10)$.

1. Use the equality of consecutive masses

For a positive Poisson mean,

$$\frac{P(X = 9)}{P(X = 8)} = \frac{\lambda}{9}.$$

Since the probabilities are equal, the ratio is one. Therefore

$$\frac{\lambda}{9} = 1 \quad \Rightarrow \quad \boxed{\lambda = 9}.$$

Equivalently, substitute the two PMFs and cancel the positive common factor $e^{-\lambda}\lambda^8/8!$.

Boundary qualification. If a degenerate Poisson model with $\lambda = 0$ were allowed, both probabilities would be zero and the equality would also hold. The conventional positive-mean interpretation, consistent with the printed answer, is $\lambda = 9$.

2. Interpret the strict lower endpoint

Because X is an integer, $4 < X \leq 10$ includes exactly $5, 6, 7, 8, 9, 10$:

$$P(4 < X \leq 10) = \sum_{r=5}^{10} \frac{e^{-9}9^r}{r!}.$$

3. Calculate the terms and sum

r	$P(X = r)$
5	0.06072688
6	0.09109032
7	0.11711612
8	0.13175564
9	0.13175564
10	0.11858008
Sum	0.65102468

$$\boxed{P(4 < X \leq 10) = 0.65102}$$

The equal rows at 8 and 9 verify the inferred mean. You can generate each successive row with $P_{r+1} = 9P_r/(r + 1)$ instead of entering the PMF repeatedly.

Exam point. The general positive-mean result is $P(X = r) = P(X = r + 1) \Rightarrow \lambda = r + 1$. Include $r + 1$, not r, when using the recurrence.

ASSIGNMENT 4 / POISSON Q2 AND Q3

A4–P2 and P3: calls and defective records

P2: at most two calls

Source: Assignment 4, Poisson Q2.

Question. Between 2 p.m. and 4 p.m., the average number of calls per minute is 2.35. Find the chance of at most two calls in one particular minute.

Under the Poisson model, let X be the number of calls in that minute. The given rate is already per minute, so $\lambda = 2.35$; the two-hour observation window is not the requested counting interval.

$$\begin{aligned} P(X \le 2) &= P(X=0) + P(X=1) + P(X=2) \\ &= e^{-2.35}\left(\frac{2.35^0}{0!} + \frac{2.35^1}{1!} + \frac{2.35^2}{2!}\right) \\ &= e^{-2.35}(1 + 2.35 + 2.76125) \\ &= \boxed{0.58282}. \end{aligned}$$

Thus about 58.282% of comparable one-minute intervals have at most two calls under this model.

P3: defects per record

Source: Assignment 4, Poisson Q3.

Question. The number of defects on a record follows a Poisson distribution with mean 0.4. Find (i) the probability of three defects, and (ii) the percentage not available for sale when only defect-free records may be sold.

Let X be defects on one record; $X \sim \text{Pois}(0.4)$.

(i) Exactly three defects:

$$P(X=3) = e^{-0.4}\frac{0.4^3}{3!} = e^{-0.4}\frac{0.064}{6} = \boxed{0.00715}.$$

This is about 0.715%, not 0.00715%.

(ii) At least one defect makes a record unsaleable:

$$P(\text{not sold}) = P(X \ge 1) = 1 - P(X=0) = 1 - e^{-0.4} = 0.32967995.$$

$$\boxed{\text{Percentage not available for sale} = 32.967995\% \approx 32.97\%.}$$

The acceptable proportion is $e^{-0.4} = 0.67032$, and acceptable plus rejected proportions sum to one.

Exam point. A mean of 0.4 defects per record is not a 40% probability of a defective record. A record can have more than one defect; $P(X \ge 1) = 1 - e^{-0.4}$.

ASSIGNMENT 4 / POISSON Q4

A4–P4: fit a Poisson frequency table

Source: Assignment 4, Poisson Q4.

Question. In 100 dictionary pages, foreign-word counts $x = 0, 1, 2, 3, 4, 5, 6$ have observed frequencies $48, 27, 12, 7, 4, 1, 1$. Find Poisson expected frequencies and the mean and variance of the fitted distribution.

1. Estimate the mean from the observed counts

$$N = 100, \qquad \sum xf = 0 + 27 + 24 + 21 + 16 + 5 + 6 = 99.$$

$$\hat{\lambda} = \bar{x} = \frac{\sum xf}{N} = \frac{99}{100} = \boxed{0.99}.$$

Do not round this to 1 before fitting. The fitted model is $X \sim \text{Pois}(0.99)$.

2. Obtain expected frequencies

$$f_r^{\text{exp}} = 100\frac{e^{-0.99}(0.99)^r}{r!}, \qquad f_0^{\text{exp}} = 100e^{-0.99} = 37.157669.$$

Then $f_{r+1}^{\text{exp}} = 0.99 f_r^{\text{exp}}/(r+1)$; for example,

$$f_1^{\text{exp}} = 0.99(37.157669) = 36.786092, \qquad f_2^{\text{exp}} = \frac{0.99}{2}(36.786092) = 18.209116.$$

Words r	Observed	Fitted probability	Expected frequency
0	48	0.37157669	37.15767
1	27	0.36786092	36.78609
2	12	0.18209116	18.20912
3	7	0.06009008	6.00901
4	4	0.01487230	1.48723
5	1	0.00294471	0.29447
6	1	0.00048588	0.04859
≥ 7	0	0.00007826	0.00783
Total	100	1	100

The final row preserves the model's infinite support. The rounded displayed frequencies may differ minutely from 100; their unrounded total is 100.

3. Mean and variance of the fitted distribution

$$\boxed{\text{E}[X] = 0.99, \qquad \text{Var}(X) = 0.99}.$$

These are fitted-model quantities. The empirical variance using divisor 100 is $263/100 - (0.99)^2 = 1.6499$; it need not equal the fitted variance. Fitting a model does not establish that it is a good description of the data. No goodness-of-fit test is requested here.

Exam point. Expected frequencies may be fractions. Keep them unrounded for calculations, even when the observed frequencies must be whole numbers.

INFERENCE THEORY

One-sample t-tests: what is being tested?

A one-sample t-test compares a population mean μ with a specified number μ_0 when the population SD is unknown. One sample is compared with a benchmark; two independent sample means require a different procedure.

Assumptions and notation

The observations $x_1, \ldots, x_n$ should form a random independent sample. Exact small-sample inference assumes a normal population. Serious skewness or outliers can make the method unreliable for small samples. The assignments' solutions below are conditional on these assumptions where they are not explicitly established.

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n},$$
$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1} = \frac{\sum x_i^2 - (\sum x_i)^2/n}{n-1},$$
$$s = \sqrt{s^2}, \qquad SE = \frac{s}{\sqrt{n}},$$
$$t_{\text{obs}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \qquad \boxed{\nu = n - 1}.$$

Here s measures spread of individual observations; SE estimates the spread of sample means. The test statistic measures the observed mean's departure from μ_0 in standard-error units. It is dimensionless.

Why the divisor is $n - 1$

Once $\bar{x}$ is estimated, the deviations sum to zero. Only $n - 1$ of them can vary freely. Dividing their squared sum by $n - 1$ makes the usual sample variance unbiased for the population variance under independent sampling with finite variance.

Under the null and normal-sample assumptions, the statistic follows Student's t distribution with $\nu = n - 1$ degrees of freedom. This distribution is symmetric around zero and has heavier tails than the standard normal distribution; its tails become lighter as ν increases.

What a test conclusion means

A small p-value says the observed departure is unusual under the null model. It does not give the probability that the null is true. A failure to reject means insufficient evidence against the null at the chosen level, not proof of equality.

Exam point. Use the sample SD s and divide it by $\sqrt{n}$. Dividing the difference in means by s alone produces the wrong test statistic. "Small sample" is not the definition of a t-test.

THE ONE-SAMPLE TEST WORKFLOW

Choose the tail, critical value and conclusion

Choose the alternative from the question before calculating t. Use $H_0 : \mu = \mu_0$ for the boundary calculation. For a one-sided test, the full null is $\mu \leq \mu_0$ or $\mu \geq \mu_0$ as appropriate.

Wording	Alternative	Reject H_0 when
Different	$\mu \neq \mu_0$	$\lvert t_{\text{obs}}\rvert > t_{1-\alpha/2,\nu}$
Greater	$\mu > \mu_0$	$t_{\text{obs}} > t_{1-\alpha,\nu}$
Less/lower	$\mu < \mu_0$	$t_{\text{obs}} < -t_{1-\alpha,\nu}$

Here $t_{q,\nu}$ is the quantile with left-tail cumulative probability q. For $T \sim t_\nu$,

$$p_{\text{two}} = 2P(T \geq |t_{\text{obs}}|), \quad p_{\text{right}} = P(T \geq t_{\text{obs}}), \quad p_{\text{left}} = P(T \leq t_{\text{obs}}).$$

Reject when $p \leq \alpha$. Use the same tail in both approaches.

Critical values used in this guide

Degrees of freedom	$t_{0.95,\nu}$	$t_{0.975,\nu}$	$t_{0.99,\nu}$
6	1.94318	2.44691	3.14267
8	1.85955	2.30600	2.89646
9	1.83311	2.26216	2.82144
11	1.79588	2.20099	2.71808
15	1.75305	2.13145	2.60248
21	1.72074	2.07961	2.51765

A two-sided 5% test uses the 0.975 column. A right-sided 2.5% test also uses 0.975, but rejects only in the right tail. A two-sided 2% test uses 0.99.

Six steps for a complete exam answer

1. Define the population mean and state the assumptions.
2. Write H_0, H_1 and the significance level α.
3. Compute or identify $\bar{x}, s, n$; show the raw-data sums if needed.
4. Find SE, t_{obs} and $\nu = n - 1$.
5. State the rejection region and compare, or calculate the correct p-value.
6. Conclude in context: sufficient or insufficient evidence for the stated alternative.

Exam point. A negative statistic does not automatically mean rejection. Its location relative to the selected tail matters. For a two-sided test, compare its absolute value with the positive critical value.

ORIGINAL WORKED EXAMPLE 11

Raw-data t-test: a complete model answer

Question. A machine is claimed to dispense a mean of 49 mL. A random independent sample from a normal population is

$$48, 49, 50, 51, 52, 50, 49, 51, 50, 50 \text{ mL}.$$

The population SD is unknown. Test for a difference at $\alpha = 0.05$.

1. Hypotheses and tail

Let μ be the population mean dispensed volume:

$$H_0 : \mu = 49, \qquad H_1 : \mu \neq 49.$$

The alternative says "different", so this is a two-sided test.

2. Sample statistics

$$n = 10, \qquad \sum x_i = 500, \qquad \bar{x} = 500/10 = 50.$$

Deviations from 50 are $-2, -1, 0, 1, 2, 0, -1, 1, 0, 0$, giving

$$\sum (x_i - \bar{x})^2 = 4 + 1 + 0 + 1 + 4 + 0 + 1 + 1 + 0 + 0 = 12.$$

$$s^2 = \frac{12}{9} = \frac{4}{3}, \qquad s = 1.15470, \qquad SE = \frac{1.15470}{\sqrt{10}} = 0.36515.$$

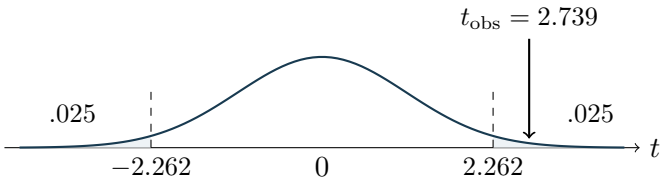
3. Test statistic and decision

$$t_{\text{obs}} = \frac{50 - 49}{0.36515} = \boxed{2.73861}, \qquad \nu = 9.$$

At $\alpha = 0.05$, reject when $|t| > t_{0.975,9} = 2.26216$. Since $2.73861 > 2.26216$, reject H_0. The two-sided p-value is approximately 0.0229, below 0.05.

4. Conclusion and graph

There is sufficient evidence at the 5% level that the population mean dispensed volume differs from 49 mL.



Schematic density; the labelled critical values and tail probabilities are from the t_9 distribution.

Exam point. Show both critical values for a two-sided test and split α equally between the two tails. The observed statistic lies in the right rejection region here.

ASSIGNMENT 6 / ONE-SAMPLE TWO-SIDED TEST

A6–Q1: mean thickness of insulating parts

Source: Assignment 6, Q1.

Question. A machine is designed to produce insulating parts with mean thickness 0.025 cm. A random sample of 10 has mean 0.024 cm and standard deviation 0.02 cm. Test the significance of the deviation.

Conventions. The question omits the significance level, so take $\alpha = 0.05$. Treat the stated SD as the sample SD and assume an independent normal sample with unknown population SD. Use the printed 0.02 cm exactly; do not substitute 0.002 cm.

1. Define the hypotheses

Let μ be the true population mean thickness:

$$H_0 : \mu = 0.025 \text{ cm}, \qquad H_1 : \mu \neq 0.025 \text{ cm}.$$

"Significance of the deviation" asks about a departure in either direction, so a two-sided test is appropriate.

2. Calculate the standard error and statistic

$$n = 10, \quad \bar{x} = 0.024, \quad s = 0.02, \quad \mu_0 = 0.025.$$

$$SE = \frac{s}{\sqrt{n}} = \frac{0.02}{\sqrt{10}} = 0.00632456 \text{ cm}.$$

$$t_{\text{obs}} = \frac{\bar{x} - \mu_0}{SE} = \frac{0.024 - 0.025}{0.00632456}$$
$$= \frac{-0.001}{0.00632456} = \boxed{-0.15811}, \qquad \nu = 9.$$

The difference is only about 0.158 standard errors below the hypothesized mean.

3. Compare with the two-sided rejection region

At $\alpha = 0.05$, reject if $t < -2.26216$ or $t > 2.26216$.

$$|t_{\text{obs}}| = 0.15811 < 2.26216 \quad \Rightarrow \quad \boxed{\text{Fail to reject } H_0}.$$

The two-sided p-value is $2P(T_9 \geq 0.15811) = 0.87786$, confirming the decision.

4. Conclude in the language of the question

At the assumed 5% significance level, the sample does not provide sufficient evidence that the machine's population mean thickness differs from 0.025 cm. The observed deviation is not statistically significant under the stated model.

The result concerns the *mean*, not whether every manufactured part is within a tolerance. The printed SD is very large relative to the difference being tested, which explains the small magnitude of t.

Exam point. Never silently repair unusual numerical data. State the given value, calculate with it, and distinguish a data concern from an arithmetic mistake.

ASSIGNMENT 6 / RAW-DATA TWO-SIDED TEST

A6–Q2: pesticide bag weights

Source: Assignment 6, Q2.

Question. The weights in kg of ten randomly selected bags are 50, 49, 52, 44, 45, 48, 46, 45, 49, 45. Test whether the average packing can be taken as 50 kg. Use $\alpha = 0.05$ because the question does not specify it; assume independent normal observations with unknown population SD.

1. Hypotheses

$$H_0 : \mu = 50 \text{ kg}, \qquad H_1 : \mu \neq 50 \text{ kg}.$$

The question tests equality, so use two tails, even though the observed mean turns out to be below 50.

2. Show the arithmetic for the sample SD

x_i	50	49	52	44	45	48	46	45	49	45
x_i^2	2500	2401	2704	1936	2025	2304	2116	2025	2401	2025

$$\sum x_i = 473, \quad \sum x_i^2 = 22437, \quad \bar{x} = 473/10 = 47.3.$$

$$\begin{aligned} SS &= \sum x_i^2 - \frac{(\sum x_i)^2}{n} = 22437 - \frac{473^2}{10} \\ &= 22437 - 22372.9 = 64.1, \\ s^2 &= \frac{64.1}{9} = 7.12222, \qquad s = 2.66875 \text{ kg}. \end{aligned}$$

3. Test statistic and degrees of freedom

$$SE = \frac{2.66875}{\sqrt{10}} = 0.84393 \text{ kg},$$

$$t_{\text{obs}} = \frac{47.3 - 50}{0.84393} = \boxed{-3.19931}, \qquad \nu = 9.$$

4. Decision and contextual conclusion

The critical values are $\pm t_{0.975,9} = \pm 2.26216$. Since

$$|-3.19931| = 3.19931 > 2.26216,$$

reject H_0. The two-sided p-value is $0.01084 < 0.05$.

At the assumed 5% level, there is sufficient evidence that the population mean packing weight differs from 50 kg. The sample indicates a lower mean, so the claim of a 50 kg average is not supported by this test.

Exam point. Use $n - 1 = 9$ in the sample variance. The denominator 10 is appropriate for the mean calculation, not for estimating variance in this t-test.

ASSIGNMENT 6 / RAW-DATA RIGHT-SIDED TEST

A6–Q4: is potato yield better than standard?

Source: Assignment 6, Q4.

Question. Twelve farms have yields 21.5, 24.5, 18.5, 17.2, 14.5, 23.2, 22.1, 20.5, 19.4, 18.1, 24.1, 18.5. Test whether the population mean yield is better than the standard 20. Use the assumed $\alpha = 0.05$ and an independent normal-sample model with unknown population SD. No yield unit is supplied.

1. Direction follows the word "better"

For yield, better means greater:

$$H_0 : \mu \le 20, \qquad H_1 : \mu > 20.$$

Calculate the test statistic at the null boundary $\mu_0 = 20$.

2. Sample calculations

x_i	x_i^2	x_i	x_i^2	x_i	x_i^2
21.5	462.25	14.5	210.25	19.4	376.36
24.5	600.25	23.2	538.24	18.1	327.61
18.5	342.25	22.1	488.41	24.1	580.81
17.2	295.84	20.5	420.25	18.5	342.25

$$\sum x_i = 242.1, \qquad \sum x_i^2 = 4984.77, \qquad \bar{x} = \frac{242.1}{12} = 20.175.$$

$$SS = 4984.77 - \frac{242.1^2}{12} = 100.4025,$$
$$s^2 = \frac{100.4025}{11} = 9.1275, \qquad s = 3.02118,$$
$$SE = \frac{3.02118}{\sqrt{12}} = 0.87214.$$

3. Statistic, critical value and decision

$$t_{\text{obs}} = \frac{20.175 - 20}{0.87214} = \boxed{0.20066}, \qquad \nu = 11.$$

Reject in the right tail if $t > t_{0.95,11} = 1.79588$. Since $0.20066 < 1.79588$, fail to reject H_0. The right-tail p-value is 0.42231.

4. Interpretation

At the assumed 5% level, there is insufficient evidence that the population mean yield exceeds 20. Although the sample mean is 0.175 above the standard, this difference is small compared with the estimated standard error of 0.87214.

Exam point. A sample mean above the benchmark does not establish improvement. Statistical significance depends on the difference relative to uncertainty, not just its sign.

ASSIGNMENT 6 / RIGHT-SIDED TEST AT 2.5%

A6–Q6: a mean greater than 70

Source: Assignment 6, Q6.

Question. Given $\bar{x} = 83$, $s = 12.5$ and $n = 22$, test a population mean of 70 against the alternative that it is greater than 70, at $\alpha = 0.025$.

Assume the observations form an independent random sample from a normal population with unknown population SD. With summary statistics only, the distribution shape and outliers cannot be checked from the provided data.

1. State the directional hypotheses

$$H_0 : \mu \leq 70, \qquad H_1 : \mu > 70.$$

Use the boundary $\mu_0 = 70$. The stated 0.025 is the probability assigned to the single right rejection tail; it should not be divided by two.

2. Calculate the standard error

$$SE = \frac{s}{\sqrt{n}} = \frac{12.5}{\sqrt{22}} = \frac{12.5}{4.69042} = 2.66501.$$

3. Standardize the departure from the null mean

$$t_{\text{obs}} = \frac{83 - 70}{2.66501} = \frac{13}{2.66501} = \boxed{4.87803}, \qquad \nu = 22 - 1 = 21.$$

The observed mean is approximately 4.88 estimated standard errors above 70.

4. Critical-value and p-value decisions

For a right-sided 2.5% test,

$$t_{1-\alpha,21} = t_{0.975,21} = 2.07961.$$

Since $4.87803 > 2.07961$, reject H_0. Equivalently,

$$p = P(T_{21} \geq 4.87803) \approx 0.0000400 < 0.025.$$

5. Conclusion

At the 2.5% significance level, there is sufficient evidence that the population mean exceeds 70.

Sketch instructions. Draw a symmetric t_{21} density, mark 0 at the centre, mark 2.07961 on the right, and shade everything to its right as the rejection region of area 0.025. Place 4.87803 well inside that shaded tail.

Table-reading check. The same numerical critical value appears in a two-sided 5% test with 21 degrees of freedom, because each of that test's tails has area 0.025. The test procedures still differ: this question rejects only for a sufficiently large positive statistic.

Exam point. Do not report the p-value as zero just because a calculator rounds it to 0.0000. Scientific notation preserves the evidence: $p \approx 4.00 \times 10^{-5}$.

ASSIGNMENT 6 / TWO-SIDED TEST AT 2%

A6–Q7: software-company P/E ratios

Source: Assignment 6, Q7.

Question. Picosoft considers a stock offering at about 13 times earnings. A random sample of seven publicly traded software firms has average P/E ratio 11.6 and sample SD 1.3. At $\alpha = 0.02$, is the population mean ratio significantly different from 13?

The printed trailing ">= 6" is not used as a separate condition. The seven observations imply six degrees of freedom. Assume independent observations from an approximately normal population; small-sample inference depends on this assumption.

1. Hypotheses

Let μ be the population mean P/E ratio for the comparison firms:

$$H_0 : \mu = 13, \qquad H_1 : \mu \neq 13.$$

The sample mean being lower does not change the stated two-sided alternative.

2. Standard error and observed statistic

$$SE = \frac{1.3}{\sqrt{7}} = 0.49135, \qquad \nu = 7 - 1 = 6.$$

$$t_{\text{obs}} = \frac{11.6 - 13}{0.49135} = \frac{-1.4}{0.49135} = \boxed{-2.84927}.$$

3. Split the significance level correctly

A two-sided 2% test allocates 0.01 to each tail:

$$t_{1-0.02/2,6} = t_{0.99,6} = 3.14267.$$

Reject only if $t < -3.14267$ or $t > 3.14267$. Here

$$|t_{\text{obs}}| = 2.84927 < 3.14267, \qquad \boxed{\text{Fail to reject } H_0}.$$

The two-sided p-value is

$$p = 2P(T_6 \geq 2.84927) = 0.02921 > 0.02.$$

4. Conclusion in context

At the 2% significance level, there is insufficient evidence that the population mean P/E ratio of the comparison firms differs from 13. This statistical result alone does not establish that a particular offering price is appropriate.

Why the chosen level matters. Since $0.02 < p < 0.05$, this same dataset would reject at a preselected 5% level, but not at the specified 2% level. Do not change α after seeing the data.

Exam point. Use the comparison mean explicitly stated in the hypothesis, 13. The background calculation $21/1.61 \approx 13.04$ explains the proposed price but does not replace the problem's null value.

ASSIGNMENT 6 / LEFT-SIDED TEST AT 5%

A6–Q9: appraised home values

Source: Assignment 6, Q9.

Question. A random sample of 12 homes has mean appraised value \$780,000 and sample SD \$49,000. Test a population mean of \$825,000 against the alternative that it is lower, at $\alpha = 0.05$.

Assume independent observations from a normal population with unknown SD. The conclusion is about the population represented by the random sample of homes.

1. Define the hypotheses and units

Work in thousands of dollars to simplify the arithmetic:

$$H_0 : \mu \geq 825, \qquad H_1 : \mu < 825.$$

Use the null boundary $\mu_0 = 825$. Dividing every monetary quantity by 1,000 leaves the statistic unchanged because both numerator and denominator scale equally.

2. Calculate the statistic

$$n = 12, \quad \bar{x} = 780, \quad s = 49, \quad \nu = 11.$$

$$SE = \frac{49}{\sqrt{12}} = 14.14508 \text{ thousand dollars.}$$

$$t_{\text{obs}} = \frac{780 - 825}{14.14508} = \frac{-45}{14.14508} = \boxed{-3.18132}.$$

3. Left-tail decision

The negative critical value is

$$-t_{0.95,11} = -1.79588.$$

Since $-3.18132 < -1.79588$, the statistic lies in the left rejection region. Reject H_0. The left-tail p-value is

$$p = P(T_{11} \leq -3.18132) = 0.00437 < 0.05.$$

4. Contextual conclusion

At the 5% level, there is sufficient evidence that the population mean appraised home value is less than \$825,000.

Sketch instructions. Draw a symmetric t_{11} density. Mark -1.79588 left of zero and shade the tail to its left, with area 0.05. Mark $t_{\text{obs}} = -3.18132$ farther left, inside the rejection region.

Common incorrect comparison. Saying "$-3.18132 < 1.79588$, so reject" is not a valid left-tail rule; nearly all negative statistics satisfy that inequality. You must compare with the *negative* critical value.

Exam point. Keep units consistent. Subtracting 780 from 825000, or using a dollar-valued numerator with a thousand-dollar-valued SD, changes the statistic by an incorrect factor.

ORIGINAL WORKED EXAMPLES 12 AND 13

Two one-sided tests: compare the logic

Example 12: evidence that the mean exceeds 100?

An independent normal sample has $n = 16$, $\bar{x} = 102$ and $s = 8$, with unknown population SD. Test for a mean greater than 100 at $\alpha = 0.05$.

$$H_0 : \mu \leq 100, \qquad H_1 : \mu > 100.$$

$$SE = \frac{8}{\sqrt{16}} = 2, \quad t_{\text{obs}} = \frac{102 - 100}{2} = 1, \quad \nu = 15.$$

The critical value is $t_{0.95,15} = 1.75305$. Because $1 < 1.75305$, fail to reject H_0. The right-tail p-value is 0.1666.

There is insufficient evidence at the 5% level that the population mean exceeds 100. The positive difference alone is not enough.

Example 13: evidence that the mean is below 50?

Nine independent observations from a normal population have $\bar{x} = 48$ and $s = 3$, with unknown population SD. Test for a mean below 50 at $\alpha = 0.05$.

$$H_0 : \mu \geq 50, \qquad H_1 : \mu < 50.$$

$$SE = \frac{3}{\sqrt{9}} = 1, \quad t_{\text{obs}} = \frac{48 - 50}{1} = -2, \quad \nu = 8.$$

The critical value is $-t_{0.95,8} = -1.85955$. Since $-2 < -1.85955$, reject H_0. The left-tail p-value is 0.0403.

There is sufficient evidence at the 5% level that the population mean is below 50.

What changes and what stays the same?

Feature	Example 12	Example 13
Alternative	Greater	Less
Rejection tail	Right	Left
Statistic	$+1$	-2
Critical boundary	$+1.75305$	-1.85955
Decision	Fail to reject	Reject

Both use $(\bar{x} - \mu_0)/(s/\sqrt{n})$ and $n - 1$ degrees of freedom. The sign and cutoff are interpreted using the alternative hypothesis.

Exam point. Never choose a one-sided alternative just because the sample mean moved in that direction. The direction should come from the question or the preselected research claim.

60 MINUTES / 50 SUGGESTED MARKS

Timed practice paper

These are additional syllabus-aligned questions written for this guide. They are not further assignment questions. Attempt all six without the solutions; the marks and time allowance are practice suggestions, not an official exam pattern.

Q1. Discrete distribution and transformation [8 marks]

$P(X = x) = c(x + 1)$ for $x = 0, 1, 2, 3$, and zero otherwise.

(a) Find c and $P(X \geq 2)$.

(b) Find $\mathrm{E}[X]$, $\mathrm{Var}(X)$ and $\mathrm{SD}(X)$.

(c) Find $\mathrm{E}[5 - 2X]$ and $\mathrm{Var}(5 - 2X)$.

Q2. Continuous distribution [9 marks]

$f(x) = kx^2$ for $0 \leq x \leq 2$, and zero otherwise.

(a) Find k and the full CDF.

(b) Calculate $P(1 < X < 1.5)$.

(c) Find the mean and variance, and determine the equal-area cutoff a.

Q3. Binomial probabilities and frequencies [8 marks]

Each of six independent products is defective with probability 0.1.

(a) Find the chance of exactly two defects and at least one defect.

(b) Give the mean and variance of the defect count.

(c) In 1,000 comparable six-product batches, how many batches are expected to contain exactly two defects?

Q4. Poisson interval conversion [8 marks]

Calls follow a Poisson process with mean four per hour. For a 30-minute interval, find the probabilities of (a) zero calls, (b) exactly two calls, and (c) at least three calls. State the count's mean and variance.

Q5. One-sample test from raw data [10 marks]

A random independent sample from a normal population is $9, 11, 10, 12, 8, 10, 11, 9, 10$. Population SD is unknown. At $\alpha = 0.05$, test whether the population mean differs from 9. Show the sample mean, variance, standard error, statistic, degrees of freedom and decision.

Q6. Explain the error [7 marks]

Correct each statement briefly: (a) $\mathrm{E}[X^2] = (\mathrm{E}[X])^2$ always; (b) a density cannot exceed 1; (c) failing to reject proves H_0; (d) three distinct matching guesses are independent Bernoulli trials; (e) for a Poisson mean 0.4, the chance of at least one event is 0.4.

Exam point. Suggested pacing: Q1–Q4 about 8 minutes each, Q5 about 15 minutes, Q6 about 8 minutes, and 5 minutes for checking endpoints, units and conclusions.

SOLUTIONS TO Q1–Q3

Practice solutions: distributions and moments

Q1. Normalize, calculate moments, transform

$$1 = c(1+2+3+4) = 10c \Rightarrow c = \frac{1}{10}, \quad P(X \geq 2) = \frac{3+4}{10} = \boxed{0.7}.$$

$$\mathrm{E}[X] = \frac{0(1)+1(2)+2(3)+3(4)}{10} = 2,$$

$$\mathrm{E}[X^2] = \frac{0+1(2)+4(3)+9(4)}{10} = 5.$$

Hence $\mathrm{Var}(X) = 5 - 2^2 = \boxed{1}$ and $\mathrm{SD}(X) = \boxed{1}$. For $Y = 5 - 2X$, linearity and the squared multiplier give

$$\mathrm{E}[Y] = 5 - 2(2) = \boxed{1}, \qquad \mathrm{Var}(Y) = (-2)^2(1) = \boxed{4}.$$

Q2. Integrate over the support $[0, 2]$

$$1 = k\int_0^2 x^2 dx = \frac{8k}{3} \Rightarrow \boxed{k = 3/8}.$$

$$F(x) = \begin{cases} 0, & x < 0, \\ x^3/8, & 0 \leq x \leq 2, \\ 1, & x > 2. \end{cases}$$

Therefore $P(1 < X < 1.5) = F(1.5) - F(1) = (3.375 - 1)/8 = \boxed{19/64 = 0.296875}$.

$$\mathrm{E}[X] = \frac{3}{8}\int_0^2 x^3 dx = \frac{3}{8}(4) = \frac{3}{2}, \quad \mathrm{E}[X^2] = \frac{3}{8}\int_0^2 x^4 dx = \frac{3}{8}\left(\frac{32}{5}\right) = \frac{12}{5}.$$

$$\mathrm{Var}(X) = \frac{12}{5} - \left(\frac{3}{2}\right)^2 = \boxed{\frac{3}{20}}.$$

For equal probability on either side, $F(a) = 1/2$, so $a^3/8 = 1/2$ and $\boxed{a = \sqrt[3]{4} = 1.58740}$. This lies in the support and is distinct from the mean of 1.5.

Q3. Binomial count per six-product batch

Let $X \sim \mathrm{Bin}(6, 0.1)$, with $q = 0.9$.

$$P(X = 2) = \binom{6}{2}(0.1)^2(0.9)^4 = 15(0.01)(0.6561) = \boxed{0.098415},$$

$$P(X \geq 1) = 1 - (0.9)^6 = \boxed{0.468559}.$$

The mean is $np = \boxed{0.6}$ and variance is $npq = \boxed{0.54}$. Among 1,000 batches, the expected number with exactly two defects is

$$1000(0.098415) = \boxed{98.415 \text{ batches}}.$$

If an approximate whole-number answer is desired, this is about 98 batches. The unrounded expectation is allowed to be nonintegral.

SOLUTIONS TO Q4–Q6

Practice solutions: Poisson and testing

Q4. Convert the rate before using Poisson

The interval is $30/60 = 1/2$ hour, so $\lambda = 4(1/2) = 2$. For $X \sim \text{Pois}(2)$,

$$P(X = 0) = e^{-2} = \boxed{0.13534},$$
$$P(X = 2) = e^{-2}\frac{2^2}{2!} = \boxed{0.27067},$$
$$P(X \geq 3) = 1 - e^{-2}\left(1 + 2 + \frac{2^2}{2!}\right) = 1 - 5e^{-2} = \boxed{0.32332}.$$

Both the mean and variance of the half-hour count equal $\boxed{2}$.

Q5. A complete two-sided decision

$H_0 : \mu = 9$ versus $H_1 : \mu \neq 9$, with $\alpha = 0.05$. There are $n = 9$ observations and their sum is 90, so $\bar{x} = 10$. Squared deviations from 10 are $1, 1, 0, 4, 4, 0, 1, 1, 0$, with sum 12.

$$s^2 = \frac{12}{8} = 1.5, \quad s = 1.22474, \quad SE = \frac{1.22474}{3} = 0.40825.$$

$$t_{\text{obs}} = \frac{10 - 9}{0.40825} = \boxed{2.44949}, \qquad \nu = 8.$$

The two-sided 5% critical values are ± 2.30600. Since $2.44949 > 2.30600$, reject H_0 (two-sided $p \approx 0.0400$). There is sufficient evidence at the 5% level that the population mean differs from 9, under the stipulated independent normal-sample assumptions.

Q6. Correct the misconceptions

(a) In general $\mathrm{E}[X^2] - (\mathrm{E}[X])^2 = \mathrm{Var}(X) \geq 0$. Equality occurs when the variance is zero, assuming finite second moments.

(b) A density can exceed 1. For example, $f(x) = 2x$ on $[0, 1]$ reaches 2, but its total area is exactly 1. Interval areas are probabilities.

(c) Failure to reject means that the evidence is insufficient at the chosen level. It does not establish that the null is true.

(d) One-to-one matching makes the guesses dependent. If two of three matches are correct, the third is forced to be correct. Enumerate the permutations.

(e) λ is an expected count, not the probability of at least one event. Here that probability is $1 - e^{-0.4} = 0.32968$.

Exam point. When reviewing errors, identify whether the failure was in the model, event translation, formula, arithmetic, or interpretation. Fix the earliest wrong step before redoing the calculation.

LAST-PASS EXAM REVISION

Final formula sheet and answer audit

DRV: $\sum p(x) = 1$, $\mathrm{E}[g(X)] = \sum g(x)p(x)$.

CRV: $\int f(x)dx = 1$, $P(a < X < b) = \int_a^b f(x)dx$,
$\mathrm{E}[g(X)] = \int g(x)f(x)dx$, $F(x) = \int_{-\infty}^{x} f(u)du$.

Moments: $\mathrm{Var}(X) = \mathrm{E}[X^2] - (\mathrm{E}[X])^2$, $\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}$.

$$\mathrm{E}[aX + b] = a\,\mathrm{E}[X] + b, \qquad \mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X).$$

Binomial: $P(X = r) = \binom{n}{r}p^r(1-p)^{n-r}$,
$\mu = np$, $\sigma^2 = np(1-p)$, $P(X \geq 1) = 1 - (1-p)^n$.

Poisson: $P(X = r) = e^{-\lambda}\lambda^r/r!$, $\mu = \sigma^2 = \lambda = \rho L$;
$P_{r+1} = \lambda P_r/(r+1)$; binomial approximation uses $\lambda = np$.

One-sample test: $s^2 = [\sum x_i^2 - (\sum x_i)^2/n]/(n-1)$,

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \qquad \nu = n - 1.$$

Two-sided: compare $|t|$ with $t_{1-\alpha/2,\nu}$. Right-sided: compare t with $t_{1-\alpha,\nu}$. Left-sided: compare t with $-t_{1-\alpha,\nu}$.

Assignment 6 decisions at the stated/assumed levels

Q	Tail	t	p	Decision
1	Two	−0.15811	0.87786	Fail to reject at 5%*
2	Two	−3.19931	0.01084	Reject at 5%*
4	Right	0.20066	0.42231	Fail to reject at 5%*
6	Right	4.87803	0.000040	Reject at 2.5%
7	Two	−2.84927	0.02921	Fail to reject at 2%
9	Left	−3.18132	0.00437	Reject at 5%

*Level assumed because the question omits it. See each worked solution for assumptions and contextual conclusions.

Before handing in your paper

- Check support, endpoint inclusion, and whether probabilities sum to one.
- Distinguish density from probability, and second moment from variance.
- Distinguish trials per experiment from repetitions of the experiment.
- For Poisson, use the mean for the actual requested interval.
- For a t-test, show hypotheses, tail, SE, degrees of freedom and a contextual conclusion; round only at the end.